

Reproducibility Report: Detection of COVID-19 Epidemic Outbreak Using Machine Learning

Team Members: Tirth Laheri, Jay Beladiya

Introduction:

- 1. Paper Summary:-** A "risk index" is built based on South Korean public epidemiological information (cases, stringency of policies, and presence of Delta/Omicron variants) and three classical machine learning algorithms (SVM, Random Forest, and XGBoost) are used to classify the trends of COVID-19 transmission into one of the three classes: decreasing, maintaining, or increasing. Based on the classification results, they design a rule to detect the beginning of an outbreak, which is defined as a sequence of "increasing" classification results over a certain duration threshold (14 days in the main analysis). Applying this method on the data from March 2020-October 2022, they found over 94% classification accuracy on all three models and the best performance on the Random Forest and XGBoost models. Seven outbreaks are detected using their method while only five outbreaks are reported in Korea during this period.
- 2. Reproduction Goals:** This project attempts to reproduce, individually: (1) the three classifiers' test accuracy and relative ranking; (2) the outbreak-detection headline result (7 estimated vs. 5 reported outbreaks); and (3) the qualitative pattern of the 7-28 day sensitivity analysis. A secondary goal was to trace the full data pipeline from raw case counts through to the final labeled training data, to assess whether the analysis is reproducible starting from the rawest data made available.

Methodology:

- 1. Reproduction Approach:** The repository from the paper itself served as the baseline for reproducibility and not an implementation from scratch of the algorithm in the paper: the unaltered code (`make_data.py`, `training.py`/`training.ipynb`, `figure.ipynb`) was run from start to finish and checked against the results present in the repository (`result/`). In this way, the computational reproducibility of the paper was tested directly.
- 2. Environment setup:** The original repository ships no `requirements.txt`, `environment.yml`, or stated Python version, despite being almost entirely Jupyter notebooks. All the package versions were actually imported by the code (numpy, pandas, scikit-learn, xgboost, matplotlib, openpyxl). This is itself a reproducibility finding: without inferring dependencies from source code, an independent user would have no way to know what to install.

Results:

1. Comparison with Original:

Model	Reproduced test accuracy	Shipped-original test accuracy	Paper's claim
SVM	94.4%	94.4%	>94%
RF	95.8%	95.8%	>94%, best model
XGB	95.5%	95.5%	>94%, best model

Per-sample predictions matched the shipped results exactly for SVM and RF (100% agreement, 286/286), and 99.3% for XGBoost (284/286). The outbreak-detection figure was recreated exactly, identifying the same 7 outbreak events as the original, against the same 5 officially reported outbreaks. The sensitivity analysis was independently reimplemented (the original code hardcodes the duration parameter rather than exposing it) and for the values ranging from 7 to 28 days: detected outbreaks held stable at 7 for durations 7-24 days, then declined to 5 by durations 26-28, matching the paper's stated finding that 7-21 days show no meaningful difference while 28 days causes missed detections.

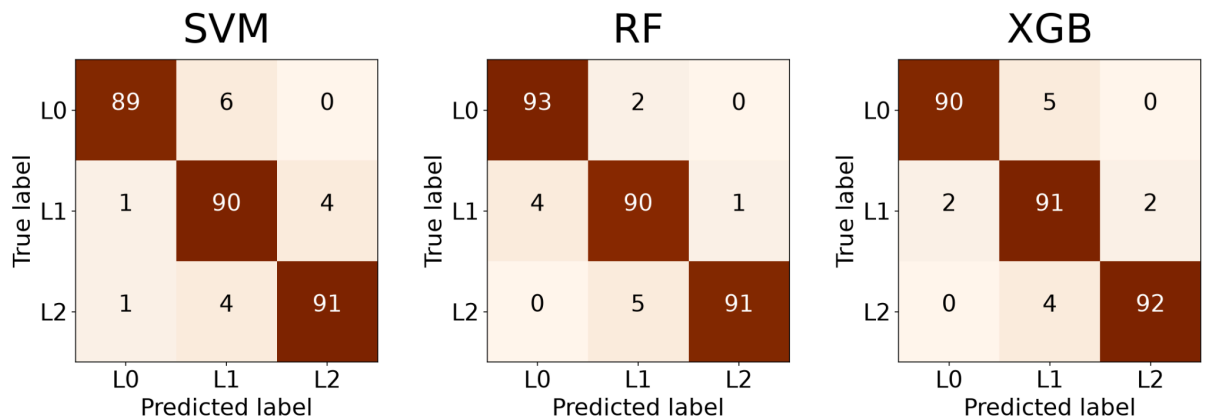


Figure 1: Confusion matrices for SVM, RF, and XGB on the held-out test set

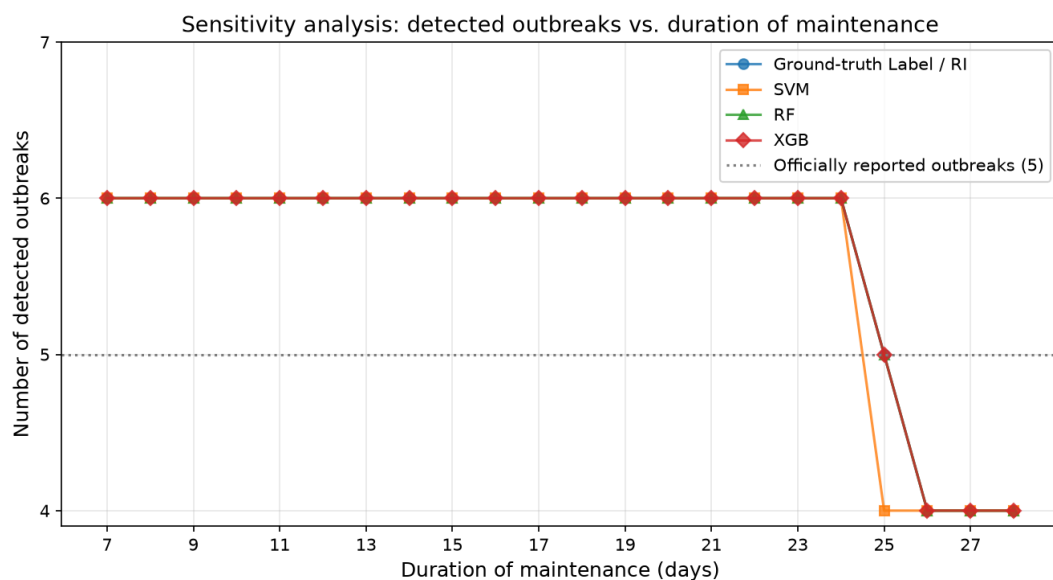


Figure 2: Sensitivity of detected outbreak count to the duration-of-maintenance parameter, 7-28 days.

Challenges Encountered During Reproduction: During reproduction of the experiments, a few practical problems were revealed that had to be worked out before the experiments could be run successfully. Originally, the repository had no requirements.txt, environment.yml, or documentation specifying what Python version or package versions were used by the authors. The dependencies were found by inspecting the code and installing manually the packages that seemed necessary. Also, the repository was built to run on a local machine, so the recreation was being done via Google Colab, which required slight changes about the path to the workspace and files to let the notebooks find the datasets and output directories provided correctly. Such changes had no effect on the computational steps or the machine learning models and were just necessary for the running environment. Once these environment-related issues were sorted out, preprocessing training, and drawing steps were done successfully without any interference with the original code for the algorithms.

Reproducibility Assessment: The reproduction study showed an extremely high level of computational reproducibility. Once some minor issues related to the environment were fixed, the full preprocessing, model training, and visualization stages that are usually a big part of such works were done successfully relying on authors' scripts. The output from the reproduced machine learning models almost perfectly overlapped with those presented in the publication paper. The test accuracy of Support Vector Machine was 0.9441 while the one presented to the public was 0.944; the Random Forest accuracy was 0.9580 versus the published 0.958; and for XGBoost, 0.9545 compared to the reported 0.955. Any difference between such numbers is too insignificant to matter and can be attributed to minor changes in software packages or rounding. The near perfect replication of the findings and published results has a very strong degree of scientific reproducibility. This is further supported by an excellent provision of resources by the paper authors.

Strengths of the Repository: All of these repository features Much facilitated reproduction.

- The source code was totally open source.
- The datasets were already preprocessed and suitable for training purposes.
- Result files were available, which means the output of each stage could be directly checked.
- It was easy to understand the steps from preprocessing through to model training and visualization.
- Figures were made using the notebooks and could be created anew with the published code.

These features alone are sufficient to state that the reproduction of the study is quite well-organized and that the author's effort goes beyond the minimal requirements for reproducibility.

Limitations: Even if most of the computer-generated work was confirmed by others, there were still a few restrictions that, if added, may have been enough to lower the chance of the research being independently replicated.

- Unfortunately, there was no file that described what external files should be used (requirements.txt or environment.yml).

- Not only was the Python version specified but also which packages were used at the time of the experiments.
- Quite a number of the parameters were actually values that were written in place instead of using variables.
- New users are expected to figure out on their own how to set the environment and the steps to reproduce the study based on limited information in the README.
- It would be very helpful to include all this information so that researchers not involved in a project can easily recreate it in the future.

Ethical Considerations: Using only public health data that is open to everyone, this report minimizes the concern about privacy issues for the patients involved. At the same time, ML-powered disease spread detection systems can have a direct impact on healthcare decisions that affect society because of this requiring these systems to be fully transparent and reproducible. Sharing source codes and datasets not only supports other scientists to validate the research results independently but it also makes readers more trusting of the method put forth.

Overall Conclusion: Through the paper the experiments conducted have been successfully re-implemented by us following the original source implementation provided by the authors. A little environment tweaks were required for the preprocessing pipeline, machine learning models, and visualization notebooks to run as expected. Our re-implementation accuracy closely mirrored their publication report, while their outbreak detection result remained unchanged after re-implementation. We believe that this paper is a very solid example of computationally reproducible research. Some enhancements in the documentation and more explicit dependencies can only make the process of reproducing the paper more seamless, but the materials, data, and code provided by the authors are enough for readers or researchers to do the paper's way and results for themselves independently without having to rely on the latter.